

Project Checkpoint #3

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Lecture summary:

- Extracellular mechanical forces influence cell structure and behaviour through integrin receptors and focal adhesions [1].
- In a process called actomyosin contractility, the cytoskeleton, especially actin filaments and myosin, generate internal contractile forces that help the cell sense and respond to its environment [1].
- Mechanical cues can lead to changes in gene expression. Actin polymerization can be caused by external stress, and releases MRTF and allows transcription factors such as YAP/TAZ to move into the nucleus which activates mechanosensitive genes [1].
- Internal changes are connected to overall cell behavior. As cells experience tension, their shape and nuclear structure adjust, influencing how genes are expressed and determining cell fate [1].

Questions:

1. When the extracellular environment is tense, integrins of the cell membrane will activate. This activation has a chain like reaction, activating another cellular component, RhoGTPase. RhoGTPase is involved in F-actin polymerization, which takes monomer G-actin and folds it into F-actin. These F-actin filaments then bind to the activated integrins and then undergo actomyosin contractility. This is where myosin works with actin filaments to bring them closer together, via the attachment and detachment of myosin heads to the actin. The contractile force that results is the responsive force to the tensile force from the ECM. This process also involves something called MRTF, which is a transcription factor. G-actin floats around the cytoplasm attached to MRTF, so when the pathway above occurs and G-actin forms F-actin, MRTF is free to enter the nucleus where it can aid in transcribing genes. Thus, the process of actin polymerization affects the expression of the mechanosensitive genes that MRTF transcribes [1].
2. TGF β is a growth factor that is held within a 'latency complex' in the ECM; it is activated and released when external stress is sensed by the ECM, thus making TGF β mechanosensitive. Activated TGF β binds to receptors on the cell surface, which then activates RhoGTPase internally. In the cell cytoplasm, G-actin monomers are bound to MRTF molecules, but active RhoGTPase separates this complex. Individual G-actin monomers undergo polymerization and 'fold into' F-actin filaments across the cell cytoplasm. Actin polymers are linked via myosin heads, which can 'pull' actin filaments inward; this is known as actomyosin contractility. When this occurs over a large network of internal F-actin, the cell itself undergoes a contractile stress [1].
3. Salari et al ran a novel experiment showing how cells tune their behavior in vitro. They placed cells on a substrate and applied an acoustic wave to a single cell. The cell

responded to the wave through oscillations, making the fluid around it move. The experiment found that if an acoustic wave is applied to different cell types, each cell responds with its own unique pattern of oscillations. When Salari et al. looked closely at the cytoskeleton, they found that stiffer cells had a greater fluid streaming velocity when an acoustic wave was applied. The main result shows an interesting connection to diseased cells. Being stiffer than healthy cells, they produce a differentiable oscillation pattern, which is a potential way to identify whether a cell is diseased or healthy [1].

4. The nucleus receives biomechanical information thanks to the LINC complex, which connects the cytoskeleton to the nucleoskeleton. As a result of this, extrinsic forces, which result in internal contractile forces through F-actin and the cytoskeleton, have a physical connection with the nucleus. Now, as the nucleus' shape changes and compresses due to external forces, where the genes/chromosomes lay and which others they are next to also change. The different "neighbourhoods" of genes/chromosomes intermingle and activate certain genes as a result, and so changing the neighbourhoods causes the transcription of different, possibly mechanosensitive, genes. The more compressed nucleus could also open the pores through which YAP/TAZ enter more widely, letting more inside the nucleus, which can also bind to/affect the transcription of specific genes [1]

Task Action Items:

- Tamar will begin planning a slideshow outline for the final presentation.
- Dyan will start brainstorming ideas for the final presentation
- Rasik will further research the osteoprotective and mechanosensitive nature of Nurr77 in relation to the RANKL-NFKB pathway
- Prabir will research affected pathways in osteoporosis.
- Abby will gather evidence from literature linking osteoporosis to lecture concepts for final presentation.
- Group will have a teams meeting over the next week to begin discussions regarding the project

References

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2. Lin S, Meng Z, Wang M, Ye Z, Long M, Zhang Y, Liu F, Chen H, Li M, Qin J, Liu H. Icariin modulates osteogenic and adipogenic differentiation in ADSCs via the Hippo-YAP/TAZ pathway: a novel therapeutic strategy for osteoporosis. *Front Pharmacol.* 2025 Jan 13;15:1510561. doi: 10.3389/fphar.2024.1510561. PMID: 39872056; PMCID: PMC11770256.
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4. Lin CY, Sassi A, Wu Y, Seaman K, Tang W, Song X, Bienenstock R, Yokota H, Sun Y, Geng F, Wang L, You L. Mechanotransduction pathways regulating YAP nuclear translocation under Yoda1 and vibration in osteocytes. *Bone.* 2025 Jan;190:117283. doi: 10.1016/j.bone.2024.117283. Epub 2024 Oct 14. PMID: 39413946.
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6. Yin H, Ruan Z, Wan T-F, et al. Metformin ameliorates osteoporosis by enhancing bone angiogenesis via the YAP1/TAZ-HIF1 α axis. *Mol Med.* 2025;31:122. doi: 10.1186/s10020-025-01169-7.

Appendix 1: Figures

Tamar:

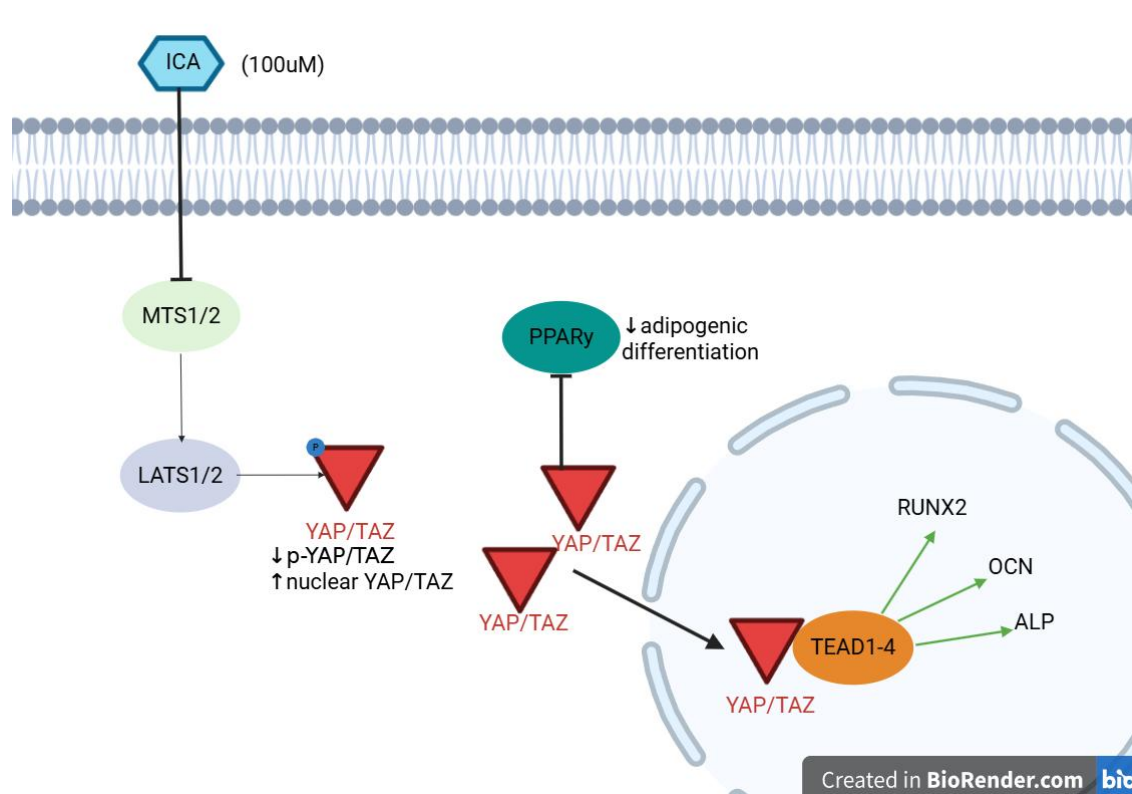


Figure 1: 100um concentration of ICA will bind to the cell membrane and inhibit the hippo pathway through downstream inhibition of hippo kinase proteins. This results in a minimal amount of phosphorylated YAP/TAZ, allowing an increased amount of nuclear YAP/TAZ to inhibit PPARγ (adipogenic differentiation protein) and migrate into the nucleus to promote osteogenic differentiation (binds to TEAD1-4). “Adapted from Lin et al., 2025 [2]

Dyan:

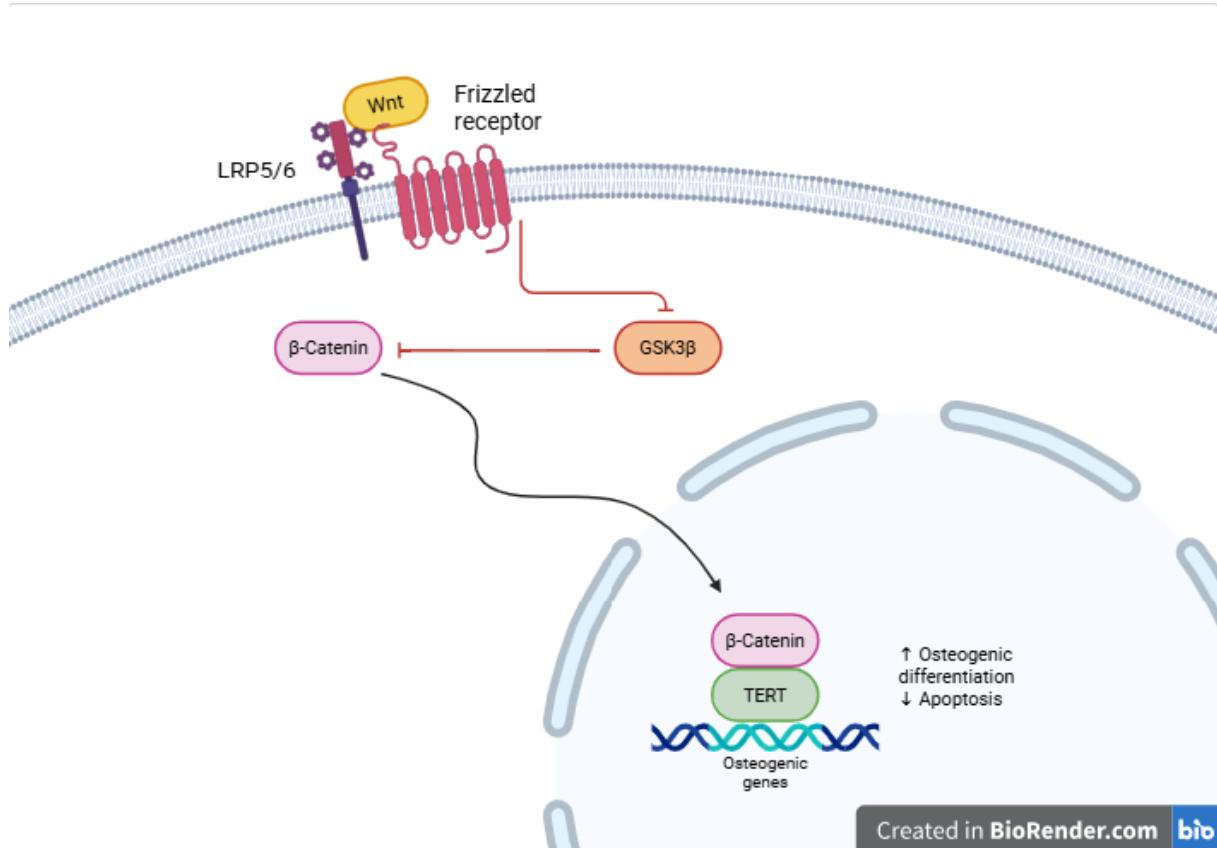


Figure 2: Activation of the Wnt/B-catenin pathway occurs when Wnt binds to the LRP5/6 and Frizzled receptors. That inhibits GSK3B, which otherwise would degrade B-catenin. An accumulation of B-catenin results in it entering the nucleus and with TERT activating osteogenic genes. Adapted from Cai et al., 2023 [3].

Prabir:

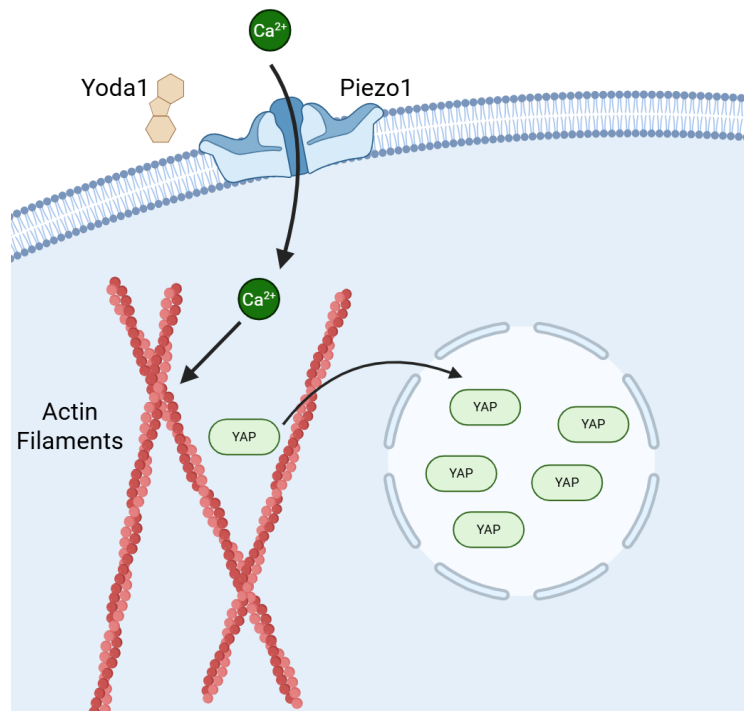


Figure 3: Yoda1 activates Piezo1 calcium channels, leading to actin filament remodeling and a less wrinkled nuclear membrane. This allows the activated YAP to enter the nucleus. Adapted from Lin et al., 2025 [4].

Rasik:

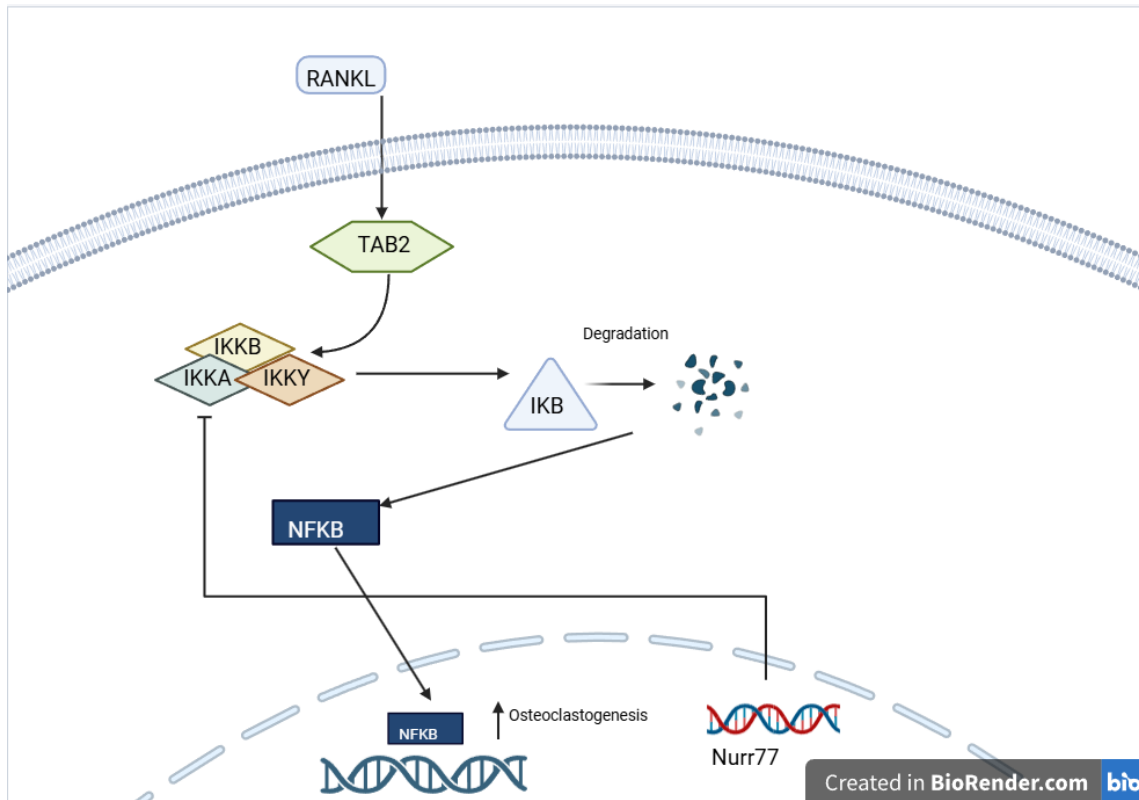


Figure 4: The RANKL-NFκB signalling pathway, which upregulates genes leading to osteoclastogenesis and its inhibition by the Nurr77 gene via inhibition of IKKα, B, and γ. Adapted from Tian et al., 2025 [5].

Appendix 2: Individual Tasks

Tamar:

- This study is both in vitro and in vivo:
 - o In vitro: Researchers conducted cell culture testing to see ICA's effects on differentiation of ADSC (adipose-derived stem cells) [2].
 - o In vivo: Survival & differentiation of ICA-treated ADSC's after transplantation [2].
- Researchers aimed to examine how different concentrations of ICA affect osteogenic and adipogenic differentiation of ADSC's. Also, they aimed to identify the specific mechanisms involved in this process, specifically in the Hippo-YAP/TAZ pathway [2].
- This study offers a potential new therapeutic treatment/strategy for osteoporosis. The efficacy of ADSCs is decreased by low post transplantation survival, however, Lin et al. study shows that ICA improves the survival capacity of transplanted stem cells. ICA has the potential to enhance bone regeneration, a significant advancement to current treatments [2].
- Researchers isolated adipose tissue from 5 healthy, 4-week old female SD rats under sterile lab conditions and retained adherent cells. ADSC cells were treated with different amounts of ICA and a control: staining and protein analysis was used to test bone & fat differentiation. The treated ADSCs were injected into 54 rats, and survival, bone formation and distribution of the cells were analyzed [2].
- ICA treated ADSCs survive at injection sites, inhibiting adipogenesis and promoting osteogenic differentiation [2].
- This results from inhibition of Hippo pathway, reducing YAP/TAZ phosphorylation and preventing YAP/TAZ from moving into the nucleus IN TURN suppressing PPARy [2].

Dyan

- This study is both in vitro and in vivo [3].
 - o In vivo ovariectomized mice (mimicking postmenopausal osteoporosis) [3].
 - o In vitro bone marrow mesenchymal stem cells (BMSC) exposed to H₂O₂ [3].
- The objective of this study was to explore how the Wnt/B-catenin pathway promotes bone marrow mesenchymal stem cells differentiation into osteoblasts and inhibits apoptosis through regulation of TERT [3].
- This study is important because it looks the dependence of molecules in one of the main pathways in bone formation [3].
- The in vivo study had 3 groups of mice, a control, OVX mice, and OVX mice with lithium chloride to activate the Wnt/B-catenin pathway [3].
- The in vitro study had H₂O₂ induced BMSCs which inhibited the pathway (and osteogenic differentiation) and then added lithium chloride to reverse the effects [3].
- They looked at the relationship between the Wnt pathway and other molecules (like TER) by introducing inhibitors [3].
- They measured the results through staining and then analyzed them with assays [3].

- The conclusion was that lithium chloride activates the Wnt/B-catenin pathway, which promotes osteogenic differentiation [3].
 - o Molecules in this pathway are interdependent, without Wnt or TERT, B-catenin cannot activate osteogenic genes [3].
- Increasing TERT expression could be a potential solution to osteoporosis [3].

Prabir

- This study was conducted in vitro using MLO-Y4 murine osteocyte-like cells [4].
- The main objective of the study was to see how Yoda1 and LMHF (low-magnitude high-frequency) vibrations can work to regulate mechanosensitive pathways and lead to the movement of YAP into the nucleus within osteocytes [4].
- This study is important as it can help provide further insights into how to better the efficacy of LMHF vibration treatments, along with identifying potential new therapeutic targets that could be used in new treatments of osteocyte-related pathologies [4].
- To test their hypotheses, the researchers planted osteocyte-like cells in vitro on tissue culture dishes treated with or without Yoda1, and a portion of the plates were subjected to LMHF vibrations. Immunofluorescence staining and microscopy were used to see/quantify the results [4].
- The first conclusion of the study was that Yoda1 and LMHF vibrations regulate the nuclear translocation of YAP through different, distinct pathways. Yoda1 activates Piezo1 calcium channels which leads to the remodeling of the actin cytoskeleton along with reduced nuclear wrinkling, which correlates with YAP moving into the nucleus (Figure 2). LMHF on the other hand uses IL6/STAT3 pathway [4].
- The other main conclusion was that Yoda1 and LMHF vibrations have a synergistic effect, with their combination resulting in the highest levels of YAP nuclear translocation, probably due to activating both pathways [4].

Rasik

- This was a study conducted both in vivo and in vitro; in vivo analysis was done through comparison of wildtype mice to Nurr77 knockout mice, while in vitro analysis was done on a tissue culture of mouse cells [5].
- The objective of the study was to determine how exactly Nurr77 prevents inflammation in people with osteoporosis and how it is osteoprotective [5].
- The study is justified because it determines if Nurr77 is medically acceptable in helping treat osteoporosis and due to its determination of Nurr77's mechanism; which allows for development of potential treatments that utilize the same mechanism [5].
- In vivo, researchers compared physiological differences between wildtype mice and Nurr77 mice via microCT scan. For in vitro analysis, researchers cultured mouse cells, using a lentivirus to under/overexpress Nurr77, and then used a Western blot to analyse key compounds involved in osteoclast differentiation [5].

- One major conclusion is that Nurr77 deletion led to increased bone loss and more osteoclast differentiation in mice, evidenced by the lower bone density and greater body fat in knockout mice and the increased presence of transcription factors associated with osteoclastogenesis [5].
- Another conclusion was the discovery that Nurr77 deletion activated the NF-KB signalling pathway in femur bone and led to inflammation, which meant that the exact cellular pathway that Nurr77 interacts with was determined, as shown in Figure 4 [5].

Abby:

- This study used in vivo experiments with mice, looking at how metformin affects bone density and blood vessel growth in osteoporotic mice; and in vitro tests on human microvascular endothelial cells (HMECs) to study molecular mechanisms [6].
- The objective was to find out if metformin can help improve bone loss in osteoporosis by promoting bone blood vessel (type H vessel) growth, and to understand the signaling pathway involved specifically the YAP/TAZ-HIF1a axis [6].
- This research is important because osteoporosis is a major health issue caused by bone loss and reduced bone vascularization, and current treatments mainly slow breakdown but don't improve blood flow or regeneration [6].
- Metformin has a possible new therapeutic role in enhancing bone healing and angiogenesis, as well as already being a safe and common diabetes drug [6].
- To test metformin's effects on bones, the study used osteoporotic mouse models (aging, ovariectomy-induced, and disuse models); bone microstructure was analyzed with micro-CT scanning and bone strength testing [6].
- Endothelial cell experiments (HMECs) were performed under normoxia and hypoxia to see how metformin affected vessel growth [6].
- Metformin increased bone mass and strength in osteoporosis models and enhanced type H blood vessel formation and stimulated angiogenesis with low oxygen [6].
- Metformin's effects work through inhibiting YAP/TAZ and activating HIF1a, showing this signaling pathway is important for bone-vessel coupling [6].